MTH 346/646 homework cover sheet

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What topics did this assignment cover? How comfortable do you feel with each of those topics?

In the course of doing this assignment, where did you get stuck? How did you get unstuck?

Is there anything you've done here that you're not confident about, that you want me to take a really close look at, or that you want me to suggest an alternate approach to?

1. Let $C : y^2 = x^3 - 49042x$. Compute the rank of C by computing the image of α and $\overline{\alpha}$ with assistance from Magma, but do not ask Magma for the rank or the Mordell-Weil group. (This is like the last example from the notes on 11/14.)

Proof. The prime factorization of 49042 is $2 \cdot 7 \cdot 31 \cdot 113$, so the divisors are

$$D = \{\pm 1, \pm 2, \pm 7, \pm 14, \pm 31, \pm 62, \pm 113, \pm 217, \pm 226, \pm 434, \\ \pm 791, \pm 1582, \pm 3503, \pm 7006, \pm 24521, \pm 49042\}$$

I used JavaScript to find the following points, given as solutions to the equation $N^2 = dM^4 - (49042/d)e^4$:

Divisor	**(M, e, N)**
1	$(1, 0, 1)$
-1	$(5289, 1999, 884489296)$
2	$(11, 1, 69)$
-2	$(13105, 1387, 178211938)$
7	$(7, 1, 99)$
-7	$(2902, 2550, 543815019)$
14	$(4, 1, 9)$
-14	$(7619, 3081, 518145689)$
31	$(86508, 4, 41667111046)$
-31	$(1, 2, 159)$
62	$(19798, 0, 3086302457)$
-62	$(1, 1, 27)$
113	$(3, 2, 47)$
-113	$(4442, 4233, 308785241)$
217	$(98892, 0, 144062911405)$
-217	$(1, 1, 3)$
226	$(1, 1, 3)$
-226	$(1279, 8908, 1168676075)$
434	$(89854, 0, 168197561503)$
-434	$(2, 3, 47)$
791	$(26243, 0, 19369356948)$
-791	$(3237, 6399, 130798511)$
1582	$(45090, 0, 80865580813)$
-1582	$(901, 10405, 601923169)$
3503	$(65544, 0, 254264630822)$
-3503	$(1, 4, 9)$
7006	$(71111, 0, 423261374961)$
-7006	$(1, 7, 99)$
24521	$(62521, 0, 612097943287)$
-24521	$(1, 11, 69)$
49042	$(37918, 0, 318401372375)$
-49042	$(2, 43, 1623).$

So in fact $\operatorname{im}\alpha = D$ and $|\operatorname{im}\alpha| = 32$.

Now for $\overline{C} : y^2 = x^3 + 196168$, we consider only positive divisors d which satisfy $N^2 = dM^4 + (196168/d)e^4$. This gives us the list

Divisor	**(M, e, N)**
1	$(1, 0, 1)$
2	$(4, 1, 314)$
4	$(1, 0, 2)$
7	$(78526, 11, 16314582762)$
8	$(2, 1, 157)$
14	$(81457, 5, 24826805419)$
28	$(82577, 5, 36082549636)$
31	$(96105, 13, 51424823882)$
56	$(81457, 10, 49653610838)$
62	$(19798, 0, 3086302457)$
113	$(1, 1, 43)$
124	$(84857, 9, 80183718050)$
217	$(98892, 0, 144062911405)$
226	$(6, 1, 542)$
248	$(19798, 0, 6172604914)$
434	$(89854, 0, 168197561503)$
452	$(1, 2, 86)$
791	$(26243, 0, 19369356948)$
868	$(98892, 0, 288125822810)$
904	$(56897, 0, 97333636911)$
1582	$(45090, 0, 80865580813)$
1736	$(89854, 0, 336395123006)$
3164	$(26243, 0, 38738713896)$
3503	$(65544, 0, 254264630822)$
6328	$(45090, 0, 161731161626)$
7006	$(71111, 0, 423261374961)$
14012	$(32772, 0, 127132315411)$
24521	$(62521, 0, 612097943287)$
28024	$(62438, 0, 652623982229)$
49042	$(37918, 0, 318401372375)$
98084	$(34993, 0, 383496529291)$
196168	$(37918, 0, 636802744750)$.

This is also of size 32, so we should have

$$2^r = \frac{|\operatorname{im}\alpha||\operatorname{im}\overline{\alpha}|}{4} = \frac{32 \cdot 32}{4} = 2^8,$$

so we get the rank $r = 8$. $\square$

2. Use Magma to determine if there is a rational point on

$$y^2 = -x^4 + 25268$$

by asking Magma to compute the rank of an elliptic curve and thinking about the images of α and $\overline{\alpha}$ for that elliptic curve.

3. $*$ This exercise goes through the proof that there are no rational points on $2y^2 = x^4 - 17$, which shows that if $C : y^2 = x^3 + 17x$, then $2 \notin \operatorname{im} \overline{\alpha}$. [I got this exercise from Nigel Boston.]

(a) Suppose that (a, b) is rational point on the curve $x^4 - 17 = 2y^2$. Show that $a = A/C$, $b = B/C^2$, where $A, B, C \in \mathbb{Z}$, $\gcd(A, B) = \gcd(A, C) = \gcd(B, C) = 1$.

(b) Rewriting the equation as $(5A^2 + 17C^2)^2 - (4B)^2 = 17(A^2 + 5C^2)^2$, show that there are $U, V \in \mathbb{Z}$ such that

$$5A^2 + 17C^2 \pm 4B = 17U^2, 5A^2 + 17C^2 \mp 4B = V^2, A^2 + 5C^2 = UV,$$

or

$$5A^2 + 17C^2 \pm 4B = 34U^2, 5A^2 + 17C^2 \mp 4B = 2V^2, A^2 + 5C^2 = 2UV.$$

(c) Hence show that (eliminating B) either

$$10A^2 + 34C^2 = 17U^2 + V^2, A^2 + 5C^2 = UV$$

or

$$5A^2 + 17C^2 = 17U^2 + V^2, A^2 + 5C^2 = 2UV.$$

(d) Show that there are no solutions to $x^2 \equiv 10 \pmod{17}$, and no solutions to $x^2 \equiv 5 \pmod{17}$ and use this to conclude that in any solution to the equations in (c) one must have $17|A$ and $17|C$. This contradicts (a).

Proof.

(a) Here's what I've got. Write $a = \alpha/\beta$ and $b = \gamma/\delta$ so that each fraction is reduced. Then

$$\frac{\alpha^4}{\beta^4} - 17 = 2 \cdot \frac{\gamma^2}{\delta^2}$$
$$\implies \alpha^4\delta^4 - 17\beta^4\delta^4 = 2\gamma^2\delta^2,$$

so we can set $A = \alpha\delta$, $C = \beta\delta$, and $B = \gamma\delta$.

(b)

$\square$